

Automated MEP Layout Synthesis and Embodied Carbon Optimization Using Multi-Agent Reinforcement Learning and Direct CAD Integration

Abstract

Context: Traditional Mechanical, Electrical, and Plumbing (MEP) drafting relies on manual polyline routing, which is time-consuming and prone to EMC clearance errors.

Methodology: We propose a Multi-Agent Reinforcement Learning (MARL) framework utilizing cooperative Q-learning agents to dynamically route power and low-voltage data circuits around architectural obstacles while maintaining strict EMC physical separation.

Results: Synthesized spatial routes are exported via a clean data contract to AutoLISP, automatically generating 90° orthogonal polylines, standard engineering symbol blocks, and a dynamic schedule table tracking circuit length and embodied carbon (kg CO2e).

1. System Architecture & Formulations

State Space: $S = [x_power, y_power, x_data, y_data]$

Reward Function: $R_total = R_distance + R_bend + R_obstacle + R_EMC$

Embodied Carbon Formula: $E_c = \sum (L_i \times EF_i)$

- Power Circuit (3x2.5mm² Cu): $EF = 0.85 \text{ kg CO2e / m}$
- Data Circuit (Cat6 UTP): $EF = 0.18 \text{ kg CO2e / m}$

2. Experimental Results Summary

Metric	Power Circuit (P1-RL)	Data Circuit (LV1-RL)	Combined System
Circuit Type	3-Drop Ring Main	Single Radial Outlet	Multi-Agent Network
Conductor / Cable	3x2.5 mm² Copper	Cat6 UTP	Mixed
Routed Length (m)	15.2 m	11.6 m	26.8 m
Embodied Carbon	12.92 kg CO2e	2.09 kg CO2e	15.01 kg CO2e
EMC Violations	0	0	Passed
Obstacle Collisions	0	0	Passed